MIL-D-0050029D(EA)
31 October 1977
SUPERSEDING
MIL-D-0050029C(MU)
26 April 1968
USED IN LIEU OF
MIL-D-50029B
15 July 1963

MILITARY SPECIFICATION

DYE, VAT YELLOW 4

This limited coordination military specification has been prepared by US Army Armament Research and Development Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-D-50029B. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers Vat Yellow 4 dye, chemically known as a dibenzpyrenequinone type dye.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
RR-S-366 - Sieve, Test.

FSC 6820

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-D-3994 - Dextrin, Technical.

STANDARDS

FEDERAL

Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Chemical and physical characteristics. Dye, Vat Yellow 4 shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Requirement	Test paragraph
Purity, percent by weight, minimum	80	4.2.4.1
Volatile matter, percent by weight, maximum	1.0	4.2.4.2
Particle size, percent by weight passing:		4.2.4.3
840-micron (No. 20) sieve, minimum	100	
297-micron (No. 50) sieve, minimum	97	
44-micron (No. 325) sieve, maximum	40	
Percent "Marcol 52" (as anti-dusting agent)	2.0 ± 0.5	4.2.4.4
3,4,8,9-Dibenzpyrene, percent by weight, maximum	0.10	4.2.4.5

3.2 Diluent. Dextrin conforming to MIL-D-3994 is acceptable as a diluent. No other diluents shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.1 Government responsibility. The Government shall perform the test for 3,4,8,9-dibenzpyrene specified in 4.2.4.5.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the Dye, Vat Yellow 4 offered for acceptance at one time which has been produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test (except for 3,4,8,9-dibenzpyrene). Three representative specimens of approximately 200 grams (g) each shall be randomly removed from the lot or batch of dye offered for acceptance and shall be placed in separate clean, dry containers labeled to identify the lot represented.

4.2.2.3 For 3,4,8,9-dibenzpyrene test. A 100-g specimen shall be removed from each sample container selected for examination in 4.2.2.1 and shall be placed in a clean, dry, wide-mouth bottle with a tight-fitting screw lid. The bottle shall be packaged in a manner suitable for shipment and shall be forwarded to the address specified (see 6.2) for testing by the Government.

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled and closed shipping container, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Polyethylene liner or closure thereof not as specified (level B only)
(d) Container closure not as specified
(e) Container damaged or leaking
(f) Desiccant not as specified or missing (level B only)
(g) Marking incorrect, missing, or illegible
(h) Palletization not as specified

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4.2.3.2 For test. Each specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4, excluding 4.2.4.5. Each specimen taken in 4.2.2.3 shall be tested by the Government as specified in 4.2.4.5. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Purity. Prepare and measure the absorbance of a standard Vat Yellow 4 dye solution (see 6.4) and a solution of the specimen as follows: Weigh to the nearest milligram (mg) approximately 0.100 g of the dye which has been previously dried to constant weight at 70° to 75° C and transfer quantitatively to a 500-milliliter (ml) volumetric flask. Dissolve in concentrated sulfuric acid and dilute to 500 ml with concentrated sulfuric acid. Mix thoroughly. Transfer a 10-ml aliquot to a 250-ml volumetric flask and dilute to 250 ml with concentrated sulfuric acid. Mix thoroughly. Transfer a portion of each solution to matched 1-centimeter cells of a spectrophotometer and measure the absorbance of the dilute solution of standard dye and the dilute solution of specimen at 568 to 571 nanometers (nm). Use the concentrated sulfuric acid used for preparing the solutions as the reference liquid in the spectrophotometer. Cell spacers or variable path length cells may be used in lieu of the prescribed dilution procedures. Calculate the percent by weight purity as follows:

$$\text{Percent purity} = \frac{ABC}{DE}$$

where: A = Percent purity of standard dye,
B = Absorbance value of specimen,
C = Weight of standard dye in grams,
D = Absorbance value of standard dye, and
E = Weight of specimen in grams.

4.2.4.2 Volatile matter. Weigh to the nearest milligram approximately 5 g of the specimen in a tared, glass-stoppered weighing bottle. Remove the stopper and dry to constant weight at 70° to 75° C. Cool to room temperature in a desiccator and weigh. Calculate the percent by weight volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100(A - B)}{W}$$

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where: A = Weight of specimen and stoppered bottle before heating in grams,
B = Weight of specimen and stoppered bottle after heating in grams, and
W = Weight of specimen in grams.

4.2.4.3 Particle size. Use sieves conforming to RR-S-366. Nest the sieves in order of decreasing mesh size (with the sieve of largest mesh on top) on a receiving pan. Weigh to the nearest 0.1 g approximately 10 g of the specimen and then use one of the following procedures:

(a) Dry method. Place the weighed specimen on the top sieve and brush the material with a camel's-hair brush until no more specimen passes through the sieve. Remove the sieve and weigh the material retained. Repeat this procedure with each of the other sieves. Calculate the percent material passing through each sieve.

(b) Wet method. Mix the weighed specimen with sufficient water in a beaker by stirring with a glass rod to produce a smooth paste. Carefully wash the thoroughly wetted specimen through the sieves. (The use of a wetting agent is permitted.) Dry the sieves in an oven at 70° to 75° C and weigh the material remaining on each sieve. Calculate the percent material passing through each sieve.

4.2.4.4 Anti-dusting agent ("Marcol 52").

(a) Reference standard. Prepare a series of six standard "Marcol 52" solutions in carbon tetrachloride containing 0.20 to 0.60 g of "Marcol 52" per 100 ml of carbon tetrachloride. (This is equivalent to 1.0 to 3.0 percent "Marcol 52" per 20 g of specimen.) Determine the absorbance of the standard solutions at 2.310 or 3.410 ± 0.002 micrometers using an infrared spectrophotometer which has been balanced at zero or at a predetermined setting using a carbon tetrachloride blank. Prepare a calibration chart of absorbance versus percent "Marcol 52" in a 20-g specimen.

(b) Preparation of specimen. Weigh 20.000 g of the specimen into a 150-ml beaker. Add 50 ml of carbon tetrachloride and agitate for at least 1/2 hour (the specimen will not dissolve). Filter with suction. Wash the funnel and precipitate with 25 to 30 ml of carbon tetrachloride. Quantitatively transfer the filtrate to a 100-ml volumetric flask and dilute to 100 ml with carbon tetrachloride.

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(c) Procedure. Determine the absorbance of the specimen solution prepared as specified in (b) at 2.310 or 3.410 ± 0.002 micrometers using the same instrument and balancing as was used in the standard solution determination. Calculate the percent "Marcol 52" from the calibration chart prepared as specified in (a). If the spectrophotometer cell thickness is such that the instrument reads off scale, further dilutions are permitted.

4.2.4.5 3,4,8,9-Dibenzpyrene. CAUTION: 3,4,8,9-Dibenzpyrene is a powerful carcinogen; adequate precautions must be taken (see 6.8). 3,4,8,9-Dibenzpyrene is light-sensitive in solution.

(a) Calibration curve. Prepare solutions of 3,4,8,9-dibenzpyrene in spectro grade cyclohexane in the concentration range of 5 to 200 nanograms per milliliter. Scan the fluorescence spectrum of each solution with a spectrofluorometer using 420 nm as the exciting wave length. Measure the intensities of the peaks at about 450 nm and plot these values against the respective concentrations to obtain the calibration curve.

(b) Procedure. Add about 10 mg of specimen (weighed to the closest 0.5 mg) and 0.2 ml of tributylamine to 10 ml of cyclohexane. Using a mechanical shaker, mix the components for at least 1 hour. Some insolubles will remain. Centrifuge the solution to clarify. Transfer the clear solution to a fluorescence cell and scan the fluorescence spectrum of the solution with the spectrofluorometer using 420 nm as the exciting wave length. The presence of 3,4,8,9-dibenzpyrene is indicated by peaks at about 450 nm and 480 nm. The 450-nm peak will be more intense. Obtain the concentration of 3,4,8,9-dibenzpyrene from the calibration curve prepared in (a). (Do not extrapolate results. If the fluorescence intensity is not on the linear part of the calibration curve, dilute the specimen solution and make new measurements.) Calculate the percent by weight 3,4,8,9-dibenzpyrene as follows:

$$\text{Percent 3,4,8,9-dibenzpyrene} = \frac{250AB}{W}$$

where: A = Concentration read from calibration curve in nanograms per milliliter
B = Milliliters of solution in which the specimen was dissolved, and
W = Weight of specimen in nanograms.

NOTE.- The "250" factor in the equation includes a correction for the percent 3,4,8,9-dibenzpyrene recovered in the extraction procedure.

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5. PACKAGING

5.1 Packing. Dye, Vat Yellow 4 shall be packed level B or commercial as specified (see 6.2).

5.1.1 Level B. Uniform quantities of no less than 68 and no more than 136 kilograms ± 1/4 percent of the marked net weight of dye shall be packed in a fiber drum conforming to rule 51, section 2 of the Uniform Freight Classification for a weight limit of over 225 but not over 300 pounds, with an aluminum barrier of 7.62 micrometers minimum thickness incorporated into one of the inner plies of the sidewall. Each drum shall be provided with a close fitting bag liner formed from polyethylene of 101.6 micrometers minimum thickness conforming to type I, class 1, grade and finish optional of L-P-378. Seams shall be completely heat sealed and shall meet the heat-seal strength requirements of L-P-378. The filled bag shall be closed by heat sealing, tying, or knotting. A minimum of eight units of desiccant conforming to MIL-D-3464 shall be placed on top of the closed liner. Drum closure shall be fully gasketed and shall form a tight seal between the rim and the top of the drum.

5.1.2 Commercial. Uniform quantities of no less than 68 and no more than 136 kilograms ± 1/4 percent of the marked net weight of dye shall be packed in accordance with Fed. Std. No. 356.

5.2 Marking. Marking for level B shipments shall be in accordance with MIL-STD-129. Marking for commercial shipments shall be in accordance with Fed. Std. No. 356. All shipments shall be marked to show lot number and date of manufacture. In addition, each container shall be durably and legibly marked with contrasting letters and background to show the net weight of contents and the following information:

CAUTION

Avoid contact with skin or clothing.
In case of contact, flush with water.
Avoid breathing dust or fumes.
Use with adequate ventilation.

5.3 Palletization. Shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 except that the pallet shall be as specified in the contract (see 6.2).

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6. NOTES

6.1 Intended use. Dye, Vat Yellow 4 is intended for use in the manufacture of colored signaling smoke mixtures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Where to send specimen for 3,4,8,9-dibenzpyrene test (see 4.2.2.3).
(c) Level of packing required (see 5.1).
(d) Type of pallet required (see 5.3).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Dye of known purity. Vat Yellow 4 dye of known purity for use in 4.2.4.1 may be obtained from Commander, US Army Armament Research and Development Command, Product Assurance Directorate, Chemical Systems Division, Aberdeen Proving Ground, MD 21010. Alternatively, the contractor may prepare his own standard dye and obtain verification of its purity from Chemical Systems Division.

6.5 Anti-dusting agent. "Marcol 52" is a highly refined petroleum oil obtainable from the Humble Oil and Refining Company.

6.6 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practice for Indicating Which Places of Figures Are To Be Considered Significant in Specified Limiting Values (ASTM E29).

6.7 Nomenclature. Indanthrene Golden Yellow GK is a typical commercial name for Vat Yellow 4 dye. It is listed in the Colour Index under number CI59100. Additional commercial names may be found on page 2430, volume 2 of the Colour Index.

6.8 3,4,8,9-Dibenzpyrene precautions. All operations and tests involving the handling of 3,4,8,9-dibenzpyrene should be in accordance with the requirements of the Occupational Safety and Health standards on carcinogens, Chapter XVII, Part 1910 of Title 29, Code of Federal Regulations.

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